Triangle Congruence: SSS, SAS, ASA, AAS and HL

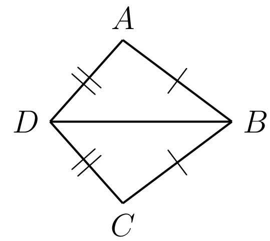
How to work:

- Matching marks mean congruent parts: one tick = one tick, two ticks = two ticks, one arc = one arc, and a small square means a right angle.
- A criterion needs **three** pairs, and *where* they sit matters. In SAS the angle is **between** the two sides; in ASA the side is **between** the two angles.
- Two arrangements are *not* criteria: SSA (two sides and a non-included angle) and AAA. HL is the one legal SSA-shaped rule, and only for right triangles.
- When you name a criterion, name the three pairs it uses. "Looks congruent" is not evidence.

1. Name the evidence. For each row, all the listed parts are congruent between $\triangle ABC$ and $\triangle DEF$. Write the criterion (SSS, SAS, ASA, AAS or HL) in the last column — or write "none" if the arrangement does not prove congruence.

	Given congruent parts	**Criterion**
(a)	$\overline{AB} \cong \overline{DE}$, $\overline{BC} \cong \overline{EF}$, $\overline{AC} \cong \overline{DF}$	
(b)	$\angle A \cong \angle D$, $\overline{AB} \cong \overline{DE}$, $\angle B \cong \angle E$	
(c)	$\overline{AB} \cong \overline{DE}$, $\angle B \cong \angle E$, $\overline{BC} \cong \overline{EF}$	
(d)	$\angle A \cong \angle D$, $\angle B \cong \angle E$, $\overline{BC} \cong \overline{EF}$	
(e)	right angles at C and F, $\overline{AB} \cong \overline{DE}$, $\overline{AC} \cong \overline{DF}$	
(f)	$\overline{AB} \cong \overline{DE}$, $\overline{BC} \cong \overline{EF}$, $\angle A \cong \angle D$	

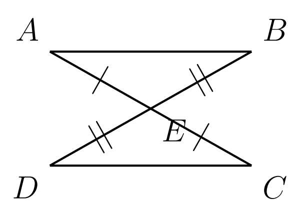
2. A shared side. In the kite below, the single ticks and the double ticks show congruent sides.



(a) $AB = 5x - 3$ and $CB = 2x + 9$: solve for x. **(b)** Which criterion proves $\triangle ABD \cong \triangle CBD$? Name the three pairs it uses, and say where the third pair comes from — it is not marked.

Answer: ________________

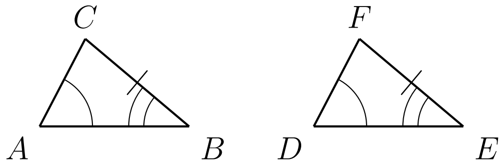
3. Vertical angles. $\overline{AC}$ and $\overline{BD}$ meet at E.



(a) $m\angle AEB = (4x + 10)^\circ$ and $m\angle CED = (6x - 20)^\circ$: solve for x. **(b)** Which criterion proves $\triangle ABE \cong \triangle CDE$? Say which pair of parts the vertical-angle theorem supplies.

Answer: ________________

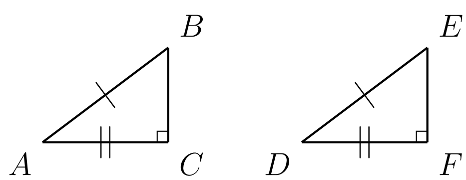
4. ASA or AAS? In $\triangle ABC$ and $\triangle DEF$ the single arcs are congruent, the double arcs are congruent, and the ticked sides $\overline{BC}$ and $\overline{EF}$ are congruent.



- $m\angle A = 52°$ and $m\angle B = 61°$. Find $m\angle C$.
- $\overline{BC}$ is not between the two marked angles. Which criterion is that?
- Now that you know the third angle, name a second criterion that also proves the triangles congruent, and explain why both work.

Answer: ________________

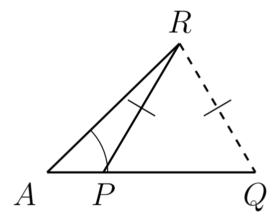
5. Hypotenuse–Leg. $\triangle ABC$ and $\triangle DEF$ have right angles at C and F. The hypotenuses are congruent and one pair of legs is congruent.



(a) The hypotenuse of $\triangle ABC$ is 13 and the marked leg $\overline{AC}$ is 5: find the length of the third side $\overline{BC}$. **(b)** Use that calculation to explain why HL really is enough — what does the Pythagorean theorem force about the unmarked legs?

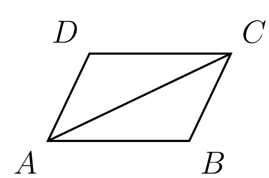
Answer: ________________

6. Why SSA is not a criterion. Both triangles below share the angle at A, the side $\overline{AR}$, and a congruent swinging side (single tick) — yet they are not congruent.



Both $\triangle ARP$ and $\triangle ARQ$ have the same angle at A, the same side $\overline{AR}$, and congruent sides $\overline{RP}$ and $\overline{RQ}$. Explain in a sentence or two why this picture shows that side-side-angle cannot be a congruence criterion, and say what extra condition turns it back into a valid rule.

7. Two-column proof. $ABCD$ is a parallelogram, so $\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$. Prove $\triangle ABC \cong \triangle CDA$.



Statements	**Reasons**
1.	1.
2.	2.
3.	3.
4.	4.

8. Judge the claim. In $\triangle PQR$ and $\triangle STU$ it is given that $\overline{PQ} \cong \overline{ST}$, $\overline{QR} \cong \overline{TU}$, and $\angle R \cong \angle U$. Rico writes "$\triangle PQR \cong \triangle STU$ by SAS".

- $m\angle R = (3x + 8)^\circ$ and $m\angle U = (5x - 16)^\circ$. Solve for x.
- Rico's arrangement is not SAS. Name it correctly and explain the difference.
- Give one extra given that would make the argument valid, and name the criterion it would then use.

Answer: ________________